

Project ARGUS (Autonomous Reconnaissance & Glide-vehicle Unification System)

Primary: Problem Statement 1 (Sensor Analysis and Integration)

Secondary: Problem Statement 2 (Edge Deployments and Drone Operation)

Project ARGUS is an autonomous aerial vehicle system designed to detect, track, and intercept highly maneuverable hypersonic cruise missiles. Traditional tracking systems struggle with hypersonic threats due to RF blackout (plasma sheathing) and unpredictable flight paths. ARGUS solves this by sequentially engaging multi-modal edge sensor data (Infrared, Electro-Optical Camera, and Air Pressure/Acoustic variations) to maintain custody of evasive targets in contested environments and optimize power. The project features a physics-based reinforcement learning simulation, a custom aerodynamic CAD design, and a physical hardware prototype demonstrating edge-compute payload integration.

Our development is divided into three distinct, interconnected phases designed to prove both the software algorithms and the hardware feasibility:

1. The Virtual Environment: Godot-Based Intercept Simulation

To train and validate our sensor-fusion and interception algorithms safely, we are building a custom simulation using the Godot engine.

The ARGUS airframe achieves multi-day mission endurance through a high-aspect-ratio, solar-glider configuration optimized for the stratosphere. By operating at altitudes of 60,000 to 90,000 feet, it maintains a superior vantage point over hypersonic cruise missiles (HCMs) transiting at lower altitudes. This platform utilizes a low-noise asymmetric sensing strategy, benefiting from the acoustic, thermal, and electronic silence of a passive glider to detect upward propagating shockwaves and thermal plumes without self-interference or detection by adversary electronic warfare suites.

Efficiency is maintained through a tiered detection architecture that employs power-gated sensor fusion. The system remains in a low-power state using passive infrasound monitoring until an anomaly triggers the Short-Wave Infrared (SWIR) array for thermal verification. Active satellite uplinks are only engaged during confirmed threats to conserve energy. This approach offers a significant cost-feasibility advantage over traditional LEO satellite constellations by utilizing commercial off-the-shelf sensors and solar-electric propulsion, allowing for the rapid deployment of persistent, scalable swarms at a fraction of the cost of orbital assets.

Adversary Modeling: We simulate hypersonic enemy targets programmed with advanced evasion heuristics (randomized glide maneuvers, erratic altitude changes).

Sensor Fusion AI: We are training models that act as the "brain" of the ARGUS drones. These agents do not receive exact target coordinates; instead, they must fuse simulated inputs from their onboard IR, optical cameras, and atmospheric pressure sensors to estimate the target's state.

2. Hardware Engineering: Aerodynamic CAD Model

To bridge the gap between software and the physical world, we are designing a custom airframe optimized for rapid deployment and sensor integration.

Sensor Placement Optimization: The CAD model will demonstrate the optimal mounting geometry for the multi-modal sensors to ensure an unoccluded, 360-degree field of view.

Aerodynamic Efficiency: The chassis is designed for stability at high speeds, ensuring that camera feeds and pressure sensors are not distorted by the drone's own wake.

3. Edge Deployment: Physical Prototype

We are building a scaled, physical prototype of the ARGUS vehicle to demonstrate real-world edge capabilities.

Live Sensor Ingestion: We will demonstrate live data ingestion from onboard cameras and environmental sensors and sensor fusion on streamed data to a macbook pro.

The vehicle will stream its data through satellite or directly to AI compute. We will use a Macbook Pro M1 Pro model as the compute hardware to run AI models. We will create a dataset of other aerial vehicle data relative to our vehicle in the simulation alongside the sensor readings and generate a dataset to train the models on. The models should be fast and low power and output to a nice, clean and cool UI.